

UF PHA 6125 — Non-Compartmental PK

Based on UF PHA 6125 Module 3

Regulatory and interpretation updates

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Module 3 2026

Reading motive

This is the UF PHA 6125 Module 3 note on non-compartmental pharmacokinetics. The course lecture presents *NCA* as a fast way to compute core *PK* metrics. My goal is to make the same material feel inevitable: once we have a concentration–time curve, *NCA* is the geometry of that curve plus a small number of elimination assumptions. It gives exposure, rate of absorption, apparent clearance, apparent volume, terminal half-life, mean residence time, and regulatory comparison metrics.

[course note]

Reader

假设读者已经读过 Module 1 and Module 2: 知道 CL controls exposure, V_d connects amount with concentration, absorption controls rate and extent of systemic input, 但还没有亲手把 sparse sampling data 变成 AUC , $C_{\max}$, λ_z , $t_{1/2}$, CL/F , V_z/F and bioequivalence metrics. This note therefore writes every formula as a small inference step from observed data.

[from curve to decision]

Update

The lecture slide deck was created before the current ICH M13A harmonized BE guideline. I keep the UF Module 3 teaching line, but update the *BE* section with FDA 2022 bioavailability guidance and FDA/ICH M13A 2024 language. One important correction: the transcript states a *BE* interval of 0.9 to 1.25, but FDA/ICH average BE language for most products is 80.00% to 125.00% for the 90% confidence interval of the geometric mean ratio [4, 2].

[2026 view]

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1 Module Overview: NCA as Curve Geometry

1.1 The official module promise

Module 3 is titled *Non-Compartmental PK*. Its promise is practical: *NCA* provides the essential *PK* information for a drug, characterizes disposition, and estimates metrics such as *AUC*, $C_{\max}$, *CL*, V_d , and terminal half-life. These metrics are then used in *BA*, *BE*, dose proportionality, food-effect, organ-impairment, formulation-bridging, and model-building work.

The lecture also frames *NCA* as a hands-on skill. Phoenix WinNonlin is mentioned because *NCA* is not only a theory topic; it is a daily clinical pharmacology workflow. A concentration table becomes an analysis object. Sampling times, units, missing values, lower-limit-of-quantification flags, terminal-point selection, and interpolation rules all become part of the scientific claim.

NCA asks: without committing to a compartmental model, what can the observed concentration–time curve itself tell us?

That phrase matters. *NCA* does not say there is no compartment. The body obviously has organs, membranes, tissues, and enzymes. It says that for many decisions, we can summarize the observed curve without fitting a named compartmental mechanism. This is why *NCA* is simultaneously modest and powerful: modest because it refuses to infer hidden states too aggressively; powerful because *AUC*, $C_{\max}$, $T_{\max}$, and $t_{1/2}$ are often exactly what a development decision needs.

1.2 Where Module 3 sits after Modules 1 and 2

Module 1 taught the big clinical pharmacology map:

dose $\longrightarrow$ exposure $\longrightarrow$ response $\longrightarrow$ decision.

Module 2 unpacked the physiological processes behind exposure:

exposure = input, distribution, and elimination acting together.

Module 3 now teaches the first standard way to measure exposure:

$$AUC = \int C(t) dt.$$

This is the conceptual handoff:

Module 1 : why exposure matters;
 Module 2 : what mechanisms shape exposure;
 Module 3 : how observed data become exposure metrics.

The order is elegant. Once *CL* and *F* are meaningful, *AUC* is not just an area:

$$AUC_{IV} = \frac{D}{CL}, \quad AUC_{PO} = \frac{FD}{CL}.$$

AUC is the empirical door into clearance and bioavailability. The curve is no longer a drawing; it is a measured balance sheet of input and removal.

1.3 A visual map of the workflow

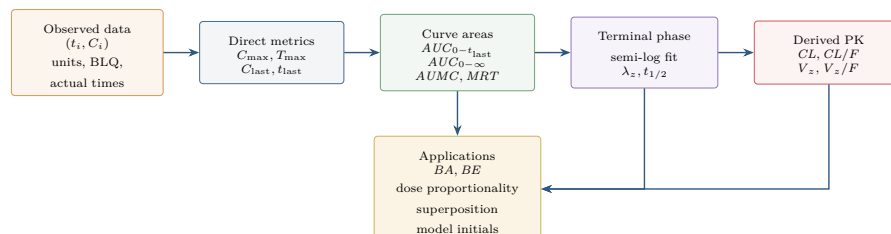


Figure 1: The *NCA* workflow. The analysis begins with observed concentrations and ends with decision-ready summaries. The terminal slope is not just another metric; it controls extrapolated area, half-life, apparent volume, and non-parametric superposition.

1.4 The two hidden assumptions

The UF lecture names two key assumptions, both about elimination:

- (i) drug is eliminated only from the central or accessible pool;
- (ii) terminal decline is mono-exponential.

The first assumption is a statement about where the measured concentration can stand in for the elimination-driving concentration. If liver or kidney clears drug from blood or plasma-accessible drug, a plasma curve is a reasonable summary. If tissue uptake, lysosomal trapping, target-mediated disposition, enterohepatic cycling, long tissue residence, or local tissue metabolism dominates, the plasma tail may be a delayed shadow rather than the actual elimination process. *NCA* can still be computed, but interpretation becomes more fragile.

The second assumption is the familiar terminal straight line:

$$C(t) = C_0 e^{-\lambda_z t} \iff \log C(t) = \log C_0 - \lambda_z t.$$

The word “terminal” is doing real work. We do not require the entire curve to be mono-exponential. Oral absorption can rise, distribution can bend, and multi-compartment movement can create curvature. What *NCA* needs for $AUC_{t_{\text{last}}-\infty}$, $t_{1/2}$, and V_z is a defensible terminal log-linear segment.

Warning. *NCA* is often called “model independent”, but that phrase should be read gently. Numerical integration of observed data is model-light. Terminal extrapolation is not model-free: it assumes a terminal exponential.

2 The Data Object

2.1 Observed samples

The raw material is a subject-level table:

$$(t_0, C_0), (t_1, C_1), \dots, (t_n, C_n), \quad 0 = t_0 < t_1 < \dots < t_n.$$

Here t_i is time after dose and C_i is measured concentration in plasma, serum, blood, urine, or another specified matrix. In most plasma *NCA* work, the table is subject-specific and treatment-period-specific. We compute individual *PK* parameters first, then summarize them across subjects.

The concentration table is not merely numbers. It carries analysis choices:

- nominal time versus actual sample time;
- units, such as ng/mL, $\mu\text{g/L}$, or mg/L;
- dose amount and route;
- assay lower limit of quantification;
- *BLQ* rules before first measurable concentration and after $T_{\max}$;
- missing samples and protocol deviations;
- whether concentration is parent drug, metabolite, total, unbound, or baseline-corrected endogenous compound.

The lecture emphasizes “write a few lines of code or filter in Excel” for $C_{\max}$ and $T_{\max}$. That is true, but the professional habit is stronger: every derived number should be reproducible from a traceable data table.

2.2 Direct curve metrics

Some *NCA* metrics are read directly from the data:

$$C_{\max} = \max_i C_i, \quad T_{\max} = t_j \quad \text{where} \quad C_j = C_{\max}.$$

$$C_{\text{last}} = C_n, \quad t_{\text{last}} = t_n,$$

where C_{last} is usually the last measurable or last quantifiable concentration, not necessarily the final scheduled sample if later samples are *BLQ*.

These four quantities already answer clinically meaningful questions:

- $C_{\max}$: peak systemic concentration;
- $T_{\max}$: observed time to peak, hence a crude input-rate marker;
- C_{last} : last observed anchor for extrapolation;
- t_{last} : how far the study followed the curve.

$T_{\max}$ deserves special caution. It is an observed order statistic, not a smooth parameter. If sampling is sparse near the peak, $T_{\max}$ can shift by design rather than biology. In *BE* work, *AUC* and $C_{\max}$ are usually primary because they are more statistically regular; $T_{\max}$ often remains descriptive unless early onset is clinically important.

2.3 A small working dataset

The following toy dataset will be used repeatedly. It is not a fitted model; it is simply a plausible concentration profile after an extravascular dose.

t (h)	0	0.5	1	2	4	6	8	12	18	24
C (mg/L)	0.00	1.10	2.35	3.10	2.40	1.65	1.15	0.58	0.22	0.085

Table 1: Toy concentration–time data for illustrating *NCA* calculations.

From the table:

$$C_{\max} = 3.10 \text{ mg/L}, \quad T_{\max} = 2 \text{ h}, \quad C_{\text{last}} = 0.085 \text{ mg/L}, \quad t_{\text{last}} = 24 \text{ h}.$$

Remark. In an actual study, the exact same formulas would be applied to each subject. Population means of concentration curves are useful for pictures, but *NCA* parameter statistics should usually be based on individual parameters, not *NCA* performed on a mean curve.

3 AUC: Exposure as Area

3.1 The exact IV bolus derivation

The cleanest *AUC* derivation comes from one-compartment *IV* bolus elimination. Suppose

$$C(t) = \frac{D}{V} e^{-(CL/V)t}.$$

Then

$$\begin{aligned} AUC_{0-\infty} &= \int_0^{\infty} C(t) \, dt \\ &= \int_0^{\infty} \frac{D}{V} e^{-(CL/V)t} \, dt \\ &= \frac{D}{V} \left[-\frac{V}{CL} e^{-(CL/V)t} \right]_0^{\infty} \\ &= \frac{D}{V} \cdot \frac{V}{CL} \\ &= \frac{D}{CL}. \end{aligned}$$

The beauty is that V cancels. *AUC* measures exposure, and for an *IV* dose under linear elimination, exposure is governed by dose divided by clearance. This is the algebraic reason Module 2 insisted on understanding CL before absorption and *BA*:

$$CL = \frac{D}{AUC_{0-\infty}} \quad \text{for IV dosing.}$$

For extravascular dosing, the systemic circulation receives only FD :

$$AUC_{0-\infty} = \frac{FD}{CL}, \quad CL/F = \frac{D}{AUC_{0-\infty}}.$$

The apparent clearance CL/F is not true clearance unless F is known.

3.2 Why numerical AUC is needed

In real studies, we do not know the exact continuous function $C(t)$. We have samples. Therefore *AUC* is split into two pieces:

$$AUC_{0-\infty} = AUC_{0-t_{\text{last}}} + AUC_{t_{\text{last}}-\infty}.$$

$AUC_{0-t_{\text{last}}}$ is estimated by interpolation between observed points. $AUC_{t_{\text{last}}-\infty}$ is extrapolated from C_{last} using the terminal slope.

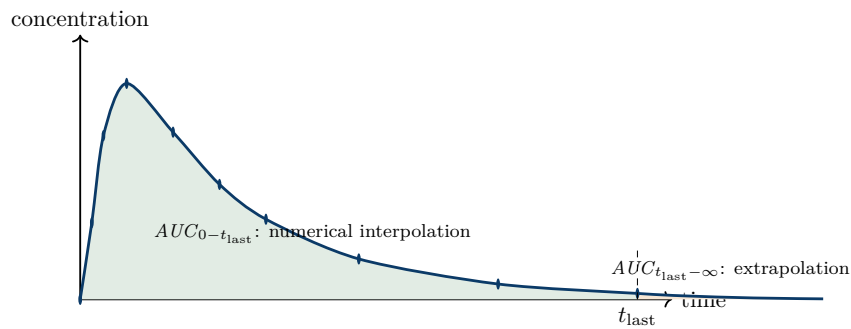


Figure 2: The two pieces of AUC in NCA . The green part is estimated from observed samples. The orange tail is model-based extrapolation.

3.3 Units

AUC has concentration times time units. If concentration is mg/L and time is hours, then

$$AUC \text{ has units } \text{mg} \cdot \text{h/L}.$$

Clearance follows by dimensional cancellation:

$$\frac{D}{AUC} = \frac{\text{mg}}{\text{mg} \cdot \text{h/L}} = \text{L/h}.$$

This is a useful self-check. If your CL output has impossible units, the bug is probably in dose units, concentration units, time units, or bioavailability interpretation.

4 Numerical Integration

4.1 Linear trapezoidal method

For two adjacent samples (t_1, C_1) and (t_2, C_2) , the linear trapezoidal area is

$$AUC_{t_1-t_2} = \frac{C_1 + C_2}{2}(t_2 - t_1).$$

The derivation is childhood geometry: average of the two parallel sides times width. Equivalently, rectangle plus triangle:

$$C_2(t_2 - t_1) + \frac{1}{2}(C_1 - C_2)(t_2 - t_1) = \frac{C_1 + C_2}{2}(t_2 - t_1).$$

Summing over all intervals gives

$$AUC_{0-t_{\text{last}}} = \sum_{i=0}^{n-1} \frac{C_i + C_{i+1}}{2}(t_{i+1} - t_i).$$

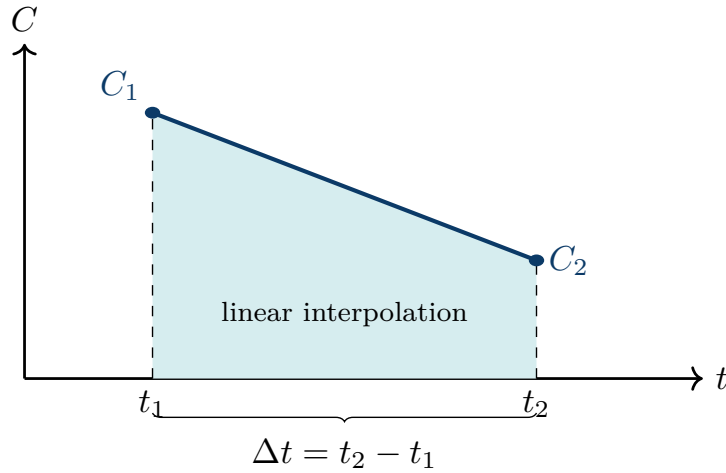


Figure 3: Linear trapezoidal area between adjacent observations. It assumes concentration changes linearly between sampled times.

The method is robust and easy. It performs especially well when sampling is dense or when the curve is close to linear over short intervals. It can overestimate area during exponential decline because the chord lies above a convex decreasing exponential curve.

4.2 Log trapezoidal method

If concentration decreases mono-exponentially between t_1 and t_2 ,

$$C(t) = C_1 e^{-k(t-t_1)} \quad \text{and} \quad C_2 = C_1 e^{-k(t_2-t_1)}.$$

Solving for k :

$$k = \frac{\log C_1 - \log C_2}{t_2 - t_1}.$$

The area is

$$\begin{aligned} AUC_{t_1-t_2} &= \int_{t_1}^{t_2} C_1 e^{-k(t-t_1)} dt \\ &= \frac{C_1 - C_2}{k} \\ &= \frac{C_1 - C_2}{\log C_1 - \log C_2} (t_2 - t_1). \end{aligned}$$

Thus the log trapezoidal formula is

$$\boxed{AUC_{t_1-t_2} = \frac{C_1 - C_2}{\log C_1 - \log C_2} (t_2 - t_1)} \quad (C_1 > C_2 > 0).$$

The assumptions are not cosmetic:

- both concentrations must be positive;
- the interval should be decreasing;
- exponential interpolation should be scientifically plausible.

If $C_1 = C_2$, the limiting value is simply $C_1(t_2 - t_1)$. If either concentration is zero or *BLQ* treated as zero, log trapezoidal interpolation is undefined.

4.3 Linear-up log-down

The practical compromise is *linear-up log-down*: use linear trapezoids when concentration rises, and log trapezoids when concentration falls.

$$AUC_{i,i+1} = \begin{cases} \frac{C_i + C_{i+1}}{2} (t_{i+1} - t_i), & C_{i+1} \geq C_i, \\ \frac{C_i - C_{i+1}}{\log C_i - \log C_{i+1}} (t_{i+1} - t_i), & C_{i+1} < C_i, \ C_i, C_{i+1} > 0. \end{cases}$$

The logic is pharmacological. During absorption and distribution, the local shape may be complicated and a linear interpolation is a neutral local approximation. During the descending log-linear portion, exponential interpolation respects first-order elimination better.

In software, there are variants: linear trapezoidal, log trapezoidal, linear-up/log-down, and rules that handle repeated rises and falls. The rule should be pre-specified for regulated analyses and reported with the results.

4.4 Numerical example

Using the toy dataset, the first interval by linear trapezoid is

$$AUC_{0-0.5} = \frac{0 + 1.10}{2} (0.5 - 0) = 0.275 \text{ mg} \cdot \text{h/L}.$$

The interval from 2 to 4 hours is decreasing. Linear trapezoidal area is

$$\frac{3.10 + 2.40}{2} (4 - 2) = 5.50.$$

Log trapezoidal area is

$$\frac{3.10 - 2.40}{\log(3.10) - \log(2.40)}(4 - 2) = 5.46.$$

The difference is small here because the interval is short and curvature mild. In sparse terminal sampling, the difference can be larger.

For the full toy table, the linear trapezoidal $AUC_{0-t_{\text{last}}}$ is approximately

$$24.38 \text{ mg} \cdot \text{h/L}.$$

The exact value will depend on the selected interpolation rule and any *BLQ* handling.

5 Terminal Slope

5.1 What lambda z means

On the terminal log-linear phase,

$$C(t) = Ae^{-\lambda_z t}.$$

Taking natural logarithms gives

$$\log C(t) = \log A - \lambda_z t.$$

Therefore, if we fit a line

$$y = a + bt, \quad y = \log C,$$

then

$$b = -\lambda_z.$$

λ_z is positive. The fitted terminal slope is negative.

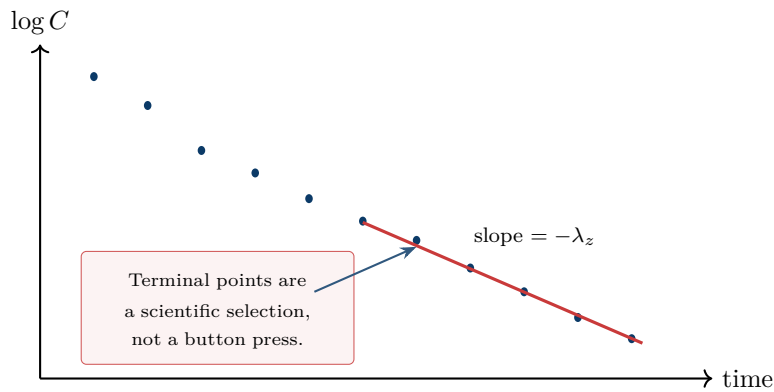


Figure 4: Terminal slope estimation on a semi-log plot. A visually straight tail supports the exponential extrapolation used by *NCA*.

5.2 Linear regression

Suppose the selected terminal points are $j = m, \dots, n$. Define

$$y_j = \log C_j.$$

The fitted line minimizes residual sum of squares:

$$(\hat{a}, \hat{b}) = \arg \min_{a, b} \sum_{j=m}^n (y_j - a - bt_j)^2.$$

Then

$$\hat{\lambda}_z = -\hat{b}.$$

The lecture introduces R^2 :

$$R^2 = 1 - \frac{\sum_j (y_j - \hat{y}_j)^2}{\sum_j (y_j - \bar{y})^2}.$$

A higher R^2 often indicates a straighter terminal segment, but it is not sufficient. Three late points can produce a high R^2 and still represent distribution, flip-flop absorption, assay noise, or a forced fit. Good *NCA* practice uses residuals, adjusted R^2 , span of the terminal phase, number of points, visual inspection, and biological context.

5.3 Terminal point selection

Terminal point selection is one of the few places in *NCA* where judgment is both unavoidable and dangerous. A defensible selection usually asks:

- Are the selected points after absorption and distribution have mostly ended?
- Do the selected points form a visually plausible straight line on a semi-log plot?
- Does adding an earlier point bend the line?
- Does removing a late noisy point change λ_z materially?
- Is the time span long enough to estimate a terminal rate, not just draw a line through adjacent noise?
- Is λ_z consistent with known physiology, earlier studies, or compartmental fits?

Warning. The last nonzero points are not automatically the terminal phase. For oral drugs with slow absorption, the terminal slope may reflect absorption rate rather than elimination rate. This is the classic flip-flop problem.

5.4 Half-life

Starting from

$$C(t) = Ae^{-\lambda_z t},$$

$t_{1/2}$ is the time required to reduce concentration by half:

$$\frac{A}{2} = Ae^{-\lambda_z t_{1/2}}.$$

Cancel A , take logs, and solve:

$$t_{1/2} = \frac{\log 2}{\lambda_z}.$$

The “five half-lives” rule follows directly:

$$\left(\frac{1}{2}\right)^5 = \frac{1}{32} \approx 3.1\%.$$

After five half-lives, about 96.9% of a mono-exponentially declining amount has disappeared from that kinetic pool. The same time-scale intuition is used for washout and time to steady state, but only when linear first-order assumptions are reasonable.

6 Extrapolation to Infinity

6.1 The tail formula

After t_{last} , assume

$$C(t) = C_{\text{last}} e^{-\lambda_z(t-t_{\text{last}})}.$$

Then

$$\begin{aligned} AUC_{t_{\text{last}}-\infty} &= \int_{t_{\text{last}}}^{\infty} C_{\text{last}} e^{-\lambda_z(t-t_{\text{last}})} dt \\ &= \frac{C_{\text{last}}}{\lambda_z}. \end{aligned}$$

Therefore

$$AUC_{0-\infty} = AUC_{0-t_{\text{last}}} + \frac{C_{\text{last}}}{\lambda_z}.$$

The extrapolated fraction is

$$\%AUC_{\text{extra}} = 100 \cdot \frac{C_{\text{last}}/\lambda_z}{AUC_{0-\infty}}.$$

Large extrapolated fraction means the total exposure estimate depends heavily on the tail assumption. The UF slide cites an older FDA BA/BE guidance recommending sampling over at least three or more terminal half-lives to capture about 90% of the relevant AUC . The current FDA 2022 BA guidance continues the same design spirit: adequate sampling is needed to characterize exposure, and the specific design depends on the purpose, dosage form, and study context [5].

6.2 AUMC tail and MRT

For mean residence time we also need the first moment curve:

$$AUMC = \int_0^{\infty} tC(t) dt.$$

The observed part is integrated numerically:

$$AUMC_{0-t_{\text{last}}} \approx \sum_i \int_{t_i}^{t_{i+1}} tC(t) dt,$$

with formulas depending on the same interpolation choice.

The extrapolated tail has a closed form:

$$\begin{aligned} AUMC_{t_{\text{last}}-\infty} &= \int_{t_{\text{last}}}^{\infty} t C_{\text{last}} e^{-\lambda_z(t-t_{\text{last}})} dt \\ &= \frac{t_{\text{last}} C_{\text{last}}}{\lambda_z} + \frac{C_{\text{last}}}{\lambda_z^2}. \end{aligned}$$

Thus

$$AUMC_{0-\infty} = AUMC_{0-t_{\text{last}}} + \frac{t_{\text{last}} C_{\text{last}}}{\lambda_z} + \frac{C_{\text{last}}}{\lambda_z^2}.$$

Mean residence time is

$$MRT = \frac{AUMC_{0-\infty}}{AUC_{0-\infty}}.$$

MRT is the average time drug molecules spend in the body or sampled kinetic system. For an *IV* bolus linear system,

$$V_{ss} = CL \cdot MRT.$$

For extravascular data, MRT includes input delay. Without additional information, it is not a pure distribution residence time.

6.3 Worked tail example

Suppose the toy profile has $C_{\text{last}} = 0.085$ mg/L, $t_{\text{last}} = 24$ h, and a terminal fit gives

$$\lambda_z = 0.17 \text{ h}^{-1}.$$

Then

$$AUC_{24-\infty} = \frac{0.085}{0.17} = 0.50 \text{ mg} \cdot \text{h/L}.$$

If $AUC_{0-t_{\text{last}}} = 24.38$, then

$$AUC_{0-\infty} = 24.88, \quad \%extra = 100 \cdot \frac{0.50}{24.88} = 2.0\%.$$

This is a comfortable extrapolated fraction. If instead C_{last} were much higher or λ_z much smaller, the same sampling schedule might leave a large part of exposure to assumption.

7 Clearance, Volume, and Residence

7.1 Clearance from AUC

For an *IV* dose:

$$CL = \frac{D}{AUC_{0-\infty}}.$$

For an extravascular dose:

$$CL/F = \frac{D}{AUC_{0-\infty}}.$$

This is one of the most common interpretation errors in early *PK* learning. Oral *NCA* does not identify *CL* and *F* separately. A high *CL/F* can mean high true clearance, low bioavailability, or both.

The Module 2 lesson should remain in view:

$$AUC_{PO} = \frac{FD}{CL}.$$

If oral exposure is low, *NCA* alone cannot tell whether the culprit is dissolution, permeability, gut metabolism, hepatic first pass, or systemic clearance. It can tell us the symptom with precision.

7.2 Volume from terminal slope

In a one-compartment terminal relationship,

$$\lambda_z = \frac{CL}{V_z}.$$

Therefore

$$V_z = \frac{CL}{\lambda_z} \quad \text{or} \quad V_z/F = \frac{CL/F}{\lambda_z} \quad \text{after extravascular dosing.}$$

Substituting $CL/F = D/AUC_{0-\infty}$ gives

$$V_z/F = \frac{D}{\lambda_z AUC_{0-\infty}}.$$

V_z is a terminal-phase apparent volume, not necessarily V_{ss} . It mixes clearance and terminal slope. In multi-compartment systems, a slow terminal slope caused by tissue redistribution can make V_z large even when steady-state distribution has a different meaning.

7.3 The Rowland-Tozer warning

Rowland and Tozer's primary-versus-secondary distinction is useful here. $t_{1/2}$, *AUC*, and V_z are often derived quantities. They are not always independent mechanistic knobs. For an *IV* bolus:

$$t_{1/2} = \frac{\log 2}{\lambda_z}, \quad \lambda_z = \frac{CL}{V_z}.$$

So changing $t_{1/2}$ can reflect changing clearance, changing distribution, or both. Similarly, AUC changes can reflect clearance or bioavailability depending on route.

The discipline is to interpret downstream *NCA* parameters through upstream physiology:

AUC	$\leftarrow$	$CL, F, \text{dose};$
$C_{\max}, T_{\max}$	$\leftarrow$	absorption rate, distribution, clearance, sampling design;
$t_{1/2}$	$\leftarrow$	CL and terminal volume;
V_z	$\leftarrow$	CL and terminal slope;
MRT	$\leftarrow$	residence in body, plus input delay when extravascular.

8 Bioavailability and Bioequivalence

8.1 Absolute and relative bioavailability

Absolute bioavailability compares extravascular exposure with intravenous exposure:

$$F_{\text{abs}} = \frac{AUC_{PO}/D_{PO}}{AUC_{IV}/D_{IV}}.$$

If written as a percentage:

$$F_{\text{abs}}(\%) = 100 \cdot \frac{AUC_{PO}D_{IV}}{AUC_{IV}D_{PO}}.$$

Relative bioavailability compares two non-*IV* products or conditions:

$$F_{\text{rel}} = \frac{AUC_{\text{test}}/D_{\text{test}}}{AUC_{\text{reference}}/D_{\text{reference}}}.$$

This is the natural language for formulation bridging, food effect, route changes, and comparing clinical-trial formulation with the to-be-marketed formulation. FDA’s 2022 BA guidance emphasizes that BA data inform safety, effectiveness, dose proportionality, food effect, and formulation life-cycle decisions [5].

8.2 Bioequivalence as a statistical statement

BE is not “two curves look similar”. It is a statistical claim about rate and extent of absorption, usually based on log-transformed *AUC* and *C*_{max}.

For most immediate-release systemic products, the modern average *BE* statement is:

$$90\% \text{ CI for } \frac{\text{geometric mean of test}}{\text{geometric mean of reference}} \subset [0.8000, 1.2500].$$

FDA’s 2001 guidance describes the average *BE* approach as a 90% confidence interval for the ratio of population geometric means, usually with 80–125% limits [4]. FDA/ICH M13A 2024 likewise states that, for the majority of drug products, primary *PK* parameters include *C*_{max} and *AUC*_{0–t} in single-dose studies, and their 90% confidence interval should lie within 80.00–125.00% [2].

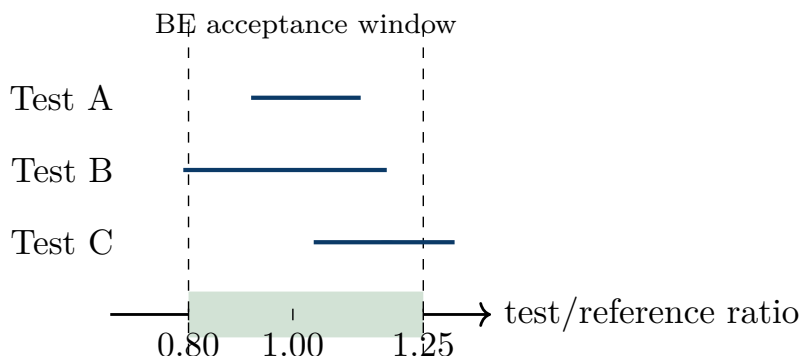


Figure 5: A schematic *BE* plot. Test A passes because the full 90% CI lies in 0.80–1.25. Test B and Test C fail because one confidence bound crosses the window, even if the point estimate looks close.

The metoprolol extended-release case discussed in the UF slide deck illustrates why *NCA* metrics matter in formulation evaluation: release-rate changes can leave one metric acceptable while another exposes a clinically relevant difference [3]. For modified-release products, shape, partial *AUC*, and dosing interval behavior may matter beyond total *AUC* and $C_{\max}$.

8.3 M13A updates

ICH M13A is important because it harmonizes scientific and technical expectations for *BE* studies of orally administered immediate-release solid oral dosage forms. Three points are especially relevant for Module 3:

- **Primary parameters.** For most single-dose *BE* studies, AUC_{0-t} and $C_{\max}$ are the core parameters. $AUC_{0-\infty}$, $T_{\max}$, λ_z , and $t_{1/2}$ are additional descriptors.
- **Sampling.** The sampling schedule should capture the concentration–time profile sufficiently for *AUC* and $C_{\max}$, with special consideration for early exposure, long half-life drugs, and partial *AUC* when onset matters.
- **Data presentation.** Individual and mean concentration–time plots on linear and log-linear scales should be provided, and concentration data should be tabulated with deviations identified.

This means *NCA* is not a side calculation at the end of a study. It influences study design before the first subject is dosed.

9 Dose Proportionality and Linearity

9.1 What dose proportionality asks

If PK is linear and bioavailability does not change with dose, then

$$AUC \propto D, \quad C_{\max} \propto D.$$

Equivalently, dose-normalized exposure is constant:

$$\frac{AUC}{D} \approx \text{constant}, \quad \frac{C_{\max}}{D} \approx \text{constant}.$$

NCA metrics are therefore the natural response variables for dose proportionality. A common model is the power model:

$$\log Y = \alpha + \beta \log D + \epsilon,$$

where Y might be AUC or $C_{\max}$. Perfect dose proportionality corresponds to

$$\beta = 1.$$

9.2 How nonlinearity appears

Nonlinearity can enter through any Module 2 mechanism:

- saturable absorption or dissolution limits;
- saturable gut or hepatic first-pass metabolism;
- saturable renal secretion;
- concentration-dependent protein binding;
- target-mediated drug disposition;
- solubility-limited exposure at high doses;
- time-varying induction or inhibition.

NCA detects the footprint before it explains the mechanism. If AUC rises more than proportionally with dose, clearance may be decreasing, first pass may be saturating, or bioavailability may be increasing. If $C_{\max}$ rises less than proportionally while AUC remains proportional, absorption rate or formulation dissolution may be changing more than extent.

9.3 A useful diagnostic table

Observation	AUC	$C_{\max}$	First interpretation
Proportional	↑ with dose	↑ with dose	Linear exposure over tested range.
More than proportional AUC	high	high or mixed	Saturable clearance or increasing F .
Less than proportional AUC	low	low	Solubility, absorption, degradation, or reduced F .
$C_{\max}$ changes more	similar	ex- altered peak	Rate/formulation/input-rate effect.
$T_{\max}$ shifts	similar	ex- peak timing changes	Absorption timing, food, gastric emptying, release profile.

Table 2: NCA metrics as a first diagnostic for dose proportionality and formulation effects.

10 Non-Parametric Superposition

10.1 The idea

Non-parametric superposition predicts multiple-dose or steady-state concentrations from a single-dose profile without fitting a compartmental model. The idea is linear systems thinking:

multiple-dose curve = sum of shifted single-dose curves.

If the dosing interval is τ , then a simple superposition expression is

$$C_{ss}(t) = \sum_{k=0}^{\infty} C_{\text{single}}(t + k\tau), \quad 0 \leq t < \tau,$$

where the unobserved late tail is extended using λ_z .

For a pure terminal exponential, the accumulation ratio is

$$R = \frac{1}{1 - e^{-\lambda_z \tau}}.$$

This is the same geometric-series logic behind repeated first-order dosing.

10.2 Assumptions

The UF lecture states the core assumptions:

- each dose acts independently of every other dose;
- rate and extent of absorption are the same for each dosing interval;
- average systemic clearance is the same for each dosing interval;
- linear *PK* applies, meaning dose proportionality and time-invariant kinetics.

These assumptions exclude many real complications: autoinduction, time-dependent inhibition, saturable absorption, adherence variation, disease evolution, target-mediated clearance, formulation changes across doses, and circadian biology. But when the assumptions are reasonable, superposition is a beautiful bridge from single-dose *NCA* to steady-state prediction.

10.3 Visualization

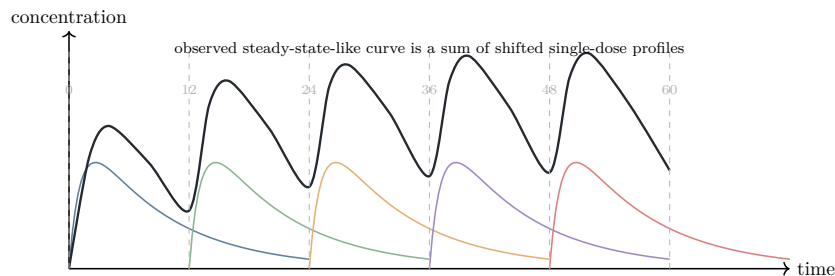


Figure 6: Schematic non-parametric superposition. Individual shifted single-dose contributions accumulate into a multiple-dose concentration curve.

11 Hands-On NCA Workflow

11.1 Before running software

Before pressing “run”, decide what the analysis means.

- (i) Identify route, dose, dosing time, matrix, analyte, units, and subject-period structure.
- (ii) Decide whether actual times or nominal times are used.
- (iii) Define *BLQ* handling before first quantifiable sample, between quantifiable samples, and after $T_{\max}$.
- (iv) Choose the interpolation rule: linear, log, or linear-up/log-down.
- (v) Pre-specify terminal slope selection rules or at least document manual selection rationale.
- (vi) Decide which parameters are primary, secondary, and exploratory.
- (vii) Check whether extravascular parameters should be reported as apparent values: CL/F , V_z/F .

This is where *NCA* resembles statistics more than arithmetic. The formulas are simple. The analysis plan protects the formulas from becoming arbitrary.

11.2 After running software

After software output, do not go straight to the parameter table. Inspect:

- individual linear-scale concentration–time plots;
- individual semi-log plots with selected terminal points;
- λ_z fit statistics and residuals;
- $\%AUC_{\text{extra}}$;
- units and dose normalization;
- impossible values such as negative half-life, $AUC_{0-\infty} < AUC_{0-t_{\text{last}}}$, or enormous apparent volume from a poor terminal slope;
- subject profiles with vomiting, missed samples, protocol deviations, or suspicious *BLQ* patterns.

For *BE* or *BA* studies, the individual curve plots are not decoration. Current FDA/ICH expectations emphasize tabulated concentration data and both individual and mean plots on linear and log-linear scales [2].

11.3 A reproducibility checklist

Item	What should be recorded
Study context	Route, formulation, dose, fasted/fed, single/multiple dose, analyte.
Data handling	Actual/nominal time, missing values, <i>BLQ</i> rule, baseline correction if endogenous.
AUC method	Linear, log, linear-up/log-down, partial <i>AUC</i> windows.
Terminal slope	Points selected, fit method, R^2 , adjusted R^2 , time span, exclusions.
Derived parameters	Whether <i>CL</i> or CL/F , V_z or V_z/F , V_{ss} only when justified.
Regulatory comparison	Log transform, <i>GMR</i> , 90% <i>CI</i> , acceptance window, primary metrics.

Table 3: Minimum reproducibility metadata for a serious *NCA* result.

12 What NCA Can and Cannot Tell You

12.1 What NCA is excellent at

NCA is excellent when the question is:

- What was systemic exposure?
- How high was the peak concentration?
- When did the observed peak occur?
- What is the apparent terminal half-life?
- What is apparent clearance or apparent volume after this route?
- Are two formulations similar in rate and extent of absorption?
- Is exposure dose proportional over the tested range?
- What initial parameter guesses might help a compartmental model?

These are not minor questions. Many label, formulation, and clinical pharmacology decisions can be made from these summaries when the study is designed well.

12.2 Where NCA becomes fragile

NCA becomes fragile when the question requires hidden mechanisms:

- separating absorption from elimination in flip-flop kinetics;
- identifying tissue compartments;
- estimating target-site concentration;
- extrapolating outside observed dose, route, formulation, or population;
- predicting untested regimens under nonlinearity;
- distinguishing low F from high CL after oral dosing;
- modeling time-varying clearance, induction, inhibition, or disease progression;
- connecting exposure to response with delays, tolerance, turnover, or feedback.

That is where compartmental PK , PK/PD , population modeling, physiologically based PK , and Bayesian inference enter. *NCA* summarizes the curve; mechanistic models explain and predict the curve.

12.3 The DiStefano distinction

DiStefano’s classic discussion of noncompartmental versus compartmental analysis is still useful because it prevents a false rivalry [1]. The choice is not “simple versus real”. It is question-dependent:

<i>NCA</i>	minimal structural assumptions, direct empirical summaries
Compartmental model	explicit structure, latent states, simulation, extrapolation

The right workflow often uses both. Run *NCA* first to understand observed exposure, half-life, and plausible starting values. Then fit a compartmental model if the scientific question demands mechanism, prediction, covariates, sparse data borrowing, or simulation.

13 Summary

13.1 One paragraph

Module 3 turns a concentration–time table into decision-ready pharmacokinetics. $C_{\max}$, $T_{\max}$, C_{last} , and t_{last} are read directly from observed data. $AUC_{0-t_{\text{last}}}$ is numerical geometry, usually linear or linear-up/log-down trapezoidal interpolation. $AUC_{0-\infty}$ adds a terminal exponential tail $C_{\text{last}}/\lambda_z$, which makes terminal slope selection one of the key scientific judgments in *NCA*. The same terminal slope gives $t_{1/2} = \log 2/\lambda_z$, while AUC gives $CL = D/AUC$ for *IV* dosing and $CL/F = D/AUC$ for extravascular dosing. $V_z = CL/\lambda_z$, $V_{ss} = CL \cdot MRT$ for appropriate *IV* settings, and $MRT = AUMC/AUC$ connect exposure to residence. These metrics support *BA*, *BE*, dose proportionality, non-parametric superposition, and starting values for compartmental models. The central discipline is interpretation: *NCA* is empirical curve geometry plus terminal exponential assumptions, not a full mechanism of absorption, distribution, metabolism, and excretion.

13.2 Formula sheet

$$C_{\max} = \max_i C_i, \quad T_{\max} = t_i \text{ at } C_{\max}, \quad C_{\text{last}} = C_n, \quad t_{\text{last}} = t_n.$$

$$AUC_{t_i-t_{i+1}}^{\text{linear}} = \frac{C_i + C_{i+1}}{2}(t_{i+1} - t_i).$$

$$AUC_{t_i-t_{i+1}}^{\log} = \frac{C_i - C_{i+1}}{\log C_i - \log C_{i+1}}(t_{i+1} - t_i),$$

$$C_i > C_{i+1} > 0.$$

$$AUC_{0-\infty} = AUC_{0-t_{\text{last}}} + \frac{C_{\text{last}}}{\lambda_z}, \quad \%extra = 100 \cdot \frac{C_{\text{last}}/\lambda_z}{AUC_{0-\infty}}.$$

$$t_{1/2} = \frac{\log 2}{\lambda_z},$$

$$CL = \frac{D}{AUC_{0-\infty}} \quad \text{for IV},$$

$$CL/F = \frac{D}{AUC_{0-\infty}} \quad \text{for extravascular.}$$

$$V_z = \frac{CL}{\lambda_z}, \quad V_z/F = \frac{CL/F}{\lambda_z}.$$

$$AUMC_{0-\infty} = AUMC_{0-t_{\text{last}}} + \frac{t_{\text{last}} C_{\text{last}}}{\lambda_z} + \frac{C_{\text{last}}}{\lambda_z^2}, \quad MRT = \frac{AUMC_{0-\infty}}{AUC_{0-\infty}}.$$

$$F_{\text{abs}} = \frac{AUC_{PO}/D_{PO}}{AUC_{IV}/D_{IV}},$$

$$F_{\text{rel}} = \frac{AUC_{\text{test}}/D_{\text{test}}}{AUC_{\text{reference}}/D_{\text{reference}}}.$$

$$BE : \quad 90\% \text{ CI}\{GMR_{\text{test/reference}}\} \subset [0.80, 1.25]$$

for most average BE settings.

13.3 Questions I should be able to answer

- (i) **Why is AUC equal to D/CL after IV bolus dosing under linear elimination?**

For an IV bolus, the whole dose reaches the systemic circulation at time zero. Under linear elimination, clearance is constant: the instantaneous elimination rate is $CLC(t)$. Mass balance says that the dose must eventually leave the body through elimination:

$$D = \int_0^{\infty} CLC(t) dt = CL \int_0^{\infty} C(t) dt = CL \cdot AUC_{0-\infty}.$$

Therefore

$$AUC_{0-\infty} = \frac{D}{CL}.$$

This argument is not limited to a one-compartment model. It needs complete systemic input, linear time-invariant clearance, and a concentration measure that is proportional to the elimination-driving concentration.

- (ii) **Which part of AUC is numerical interpolation and which part is model-based extrapolation?**

$AUC_{0-t_{\text{last}}}$ is numerical interpolation between observed data points. We do not know the true curve between t_i and t_{i+1} , so we approximate the area by linear trapezoids, log trapezoids, or linear-up/log-down rules. By contrast,

$$AUC_{t_{\text{last}}-\infty} = \frac{C_{\text{last}}}{\lambda_z}$$

is model-based extrapolation. It assumes that after t_{last} , the concentration declines mono-exponentially with terminal slope λ_z . Thus $AUC_{0-\infty}$ is a hybrid:

$$AUC_{0-\infty} = \underbrace{AUC_{0-t_{\text{last}}}}_{\text{data-driven numerical geometry}} + \underbrace{C_{\text{last}}/\lambda_z}_{\text{terminal exponential assumption}}.$$

- (iii) **When should log trapezoidal interpolation be preferred over linear trapezoidal interpolation?**

Log trapezoidal interpolation is preferred over clearly declining positive concentration intervals when the decline is approximately exponential. In that setting, the curve is closer to a straight line on the log-concentration scale than on the concentration scale, and the linear trapezoid tends to overestimate area because the chord sits above a convex exponential decline. It should not be used across rising intervals, zero values, negative baseline-corrected concentrations, or BLQ values treated as zero. This is why many workflows use linear-up/log-down: linear during absorption or rising portions, log during post-peak decline.

- (iv) **Why is terminal slope selection not a purely automatic problem?**

λ_z is estimated from a chosen set of late points, but the computer does not know whether those points are truly terminal. A high R^2 can be obtained

from distribution-phase points, sparse noisy points, flip-flop absorption, enterohepatic recycling, assay artifacts, or a short late segment that looks straight by accident. Terminal selection therefore requires visual inspection, adjusted R^2 , residuals, number of points, time span, consistency with prior physiology, and sensitivity analysis. The question is not only “which line fits best?” but “which line represents the elimination-dominant terminal process?”

(v) **Why can oral NCA estimate CL/F but not CL alone?**

After oral or other extravascular dosing,

$$AUC_{PO} = \frac{FD}{CL}.$$

Rearranging gives

$$\frac{CL}{F} = \frac{D}{AUC_{PO}},$$

which is CL/F in the common NCA notation. The data identify the ratio CL/F , not its two components. Low exposure could mean high systemic clearance, low bioavailability, incomplete absorption, gut degradation, intestinal metabolism, hepatic first pass, or some mixture. An IV reference, mass-balance evidence, or mechanistic absorption model is needed to separate F from CL .

(vi) **Why are $C_{\max}$ and $T_{\max}$ rate-related but not pure absorption parameters?**

$C_{\max}$ and $T_{\max}$ are read from the observed systemic concentration curve. They are influenced by absorption rate, but they are also shaped by distribution, clearance, dose, formulation, sampling schedule, and assay variability. A faster absorption process often increases $C_{\max}$ and shortens $T_{\max}$, but a change in clearance or distribution can move the peak too. Therefore $C_{\max}$ and $T_{\max}$ are useful rate-sensitive summaries, not isolated measurements of the absorption rate constant.

(vii) **What does $\%AUC_{\text{extra}}$ warn us about?**

$\%AUC_{\text{extra}}$ measures how much of total exposure is inferred beyond the last measurable concentration:

$$\%AUC_{\text{extra}} = 100 \cdot \frac{C_{\text{last}}/\lambda_z}{AUC_{0-\infty}}.$$

A large value warns that $AUC_{0-\infty}$ depends heavily on the terminal slope assumption rather than observed data. It may indicate insufficient late sampling, an uncertain λ_z , a long half-life, inappropriate terminal-point selection, or a study design that cannot support reliable total exposure. In that situation, $AUC_{0-t_{\text{last}}}$ may be more defensible than $AUC_{0-\infty}$ for some comparisons.

(viii) **Why does BE use geometric mean ratios and 90% confidence intervals?**

AUC and $C_{\max}$ are positive, multiplicative, often right-skewed quantities, so BE analysis usually works on the log scale. On the log scale, treatment differences become differences in means; after exponentiation, they become ratios of geometric means. This matches the scientific question: not whether test and reference differ by a fixed concentration amount, but whether exposure is acceptably close in relative terms.

The 90% confidence interval comes from the two one-sided tests logic. To show average BE , we need enough evidence that the test/reference ratio is not too low and not too high. For many products, this means the 90% CI for the geometric mean ratio lies within 0.80 to 1.25. The interval is stricter than simply asking whether the point estimate is near 1: both uncertainty and bias matter.

(ix) **What assumptions make non-parametric superposition reasonable?**

Non-parametric superposition treats the single-dose concentration profile as a response kernel and stacks shifted copies of it across repeated doses. This is reasonable when PK is linear and time-invariant: dose proportionality holds, clearance and distribution do not change over time, absorption behavior is reproducible, and one dose does not alter the disposition of the next dose except through additive residual concentrations. It becomes fragile with saturable metabolism, autoinduction, enzyme inhibition, time-varying disease, formulation changes, nonlinear protein binding, target-mediated disposition, or circadian/input changes that make each dose behave differently.

(x) **When should I stop using NCA alone and move to a compartmental or population model?**

Stop using NCA alone when the question needs a mechanism, prediction, covariate explanation, sparse-data borrowing, or simulation. Examples include optimizing untested dosing regimens, estimating individual exposure from sparse clinical samples, separating absorption from elimination, explaining food or formulation effects, modeling accumulation under nonlinear clearance, linking exposure to response, handling time-varying covariates, studying special populations, or making pediatric/renal/hepatic-impairment predictions. NCA is excellent for summarizing observed curves; compartmental and population models are needed when we must explain the curve, borrow information across subjects, and predict outside the observed design.

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